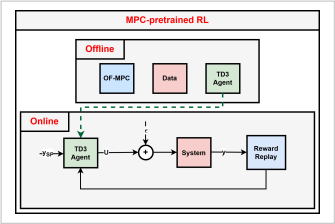
Practical Reinforcement Learning Control Framework for Complex Process Systems (Amir H. Hamedi)

Research direction

- Making reinforcement learning practical for industrial process control by keeping MPC in the loop.
- Case studies: styrene polymerization reactor and Aspen Dynamics C_2 splitter.

MPC-pretrained RL



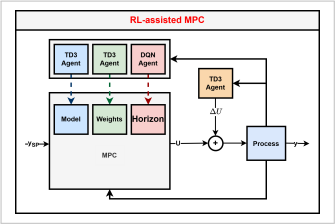
88% avg. improvement

86% avg. improvement

Polymer — lower final offset vs. previous approach

C_2 splitter — faster settling than MPC

RL-assisted MPC



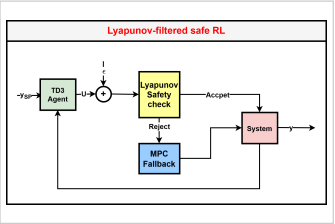
Polymer combined

+1.89 full-run reward vs MPC

Evaluation episode: +2.38

C_2: improved; final results WIP

Lyapunov-filtered safe RL



Polymer completed

RL actions certified online

C_2 splitter extension in progress